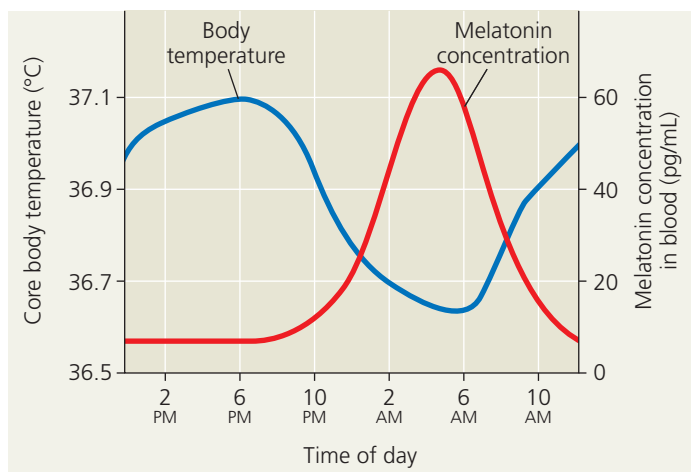
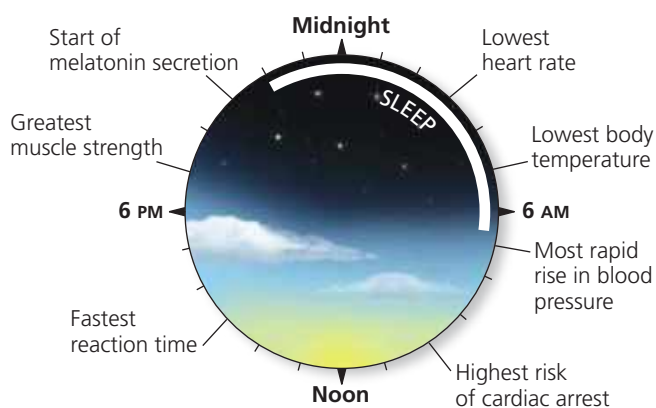


▼ **Figure 40.9 Human circadian rhythm.**



(a) Variation in core body temperature and melatonin concentration in blood. Researchers studied resting but awake volunteers in an isolation chamber with constant temperature and low light. (Melatonin is a hormone secreted by the pineal gland.)

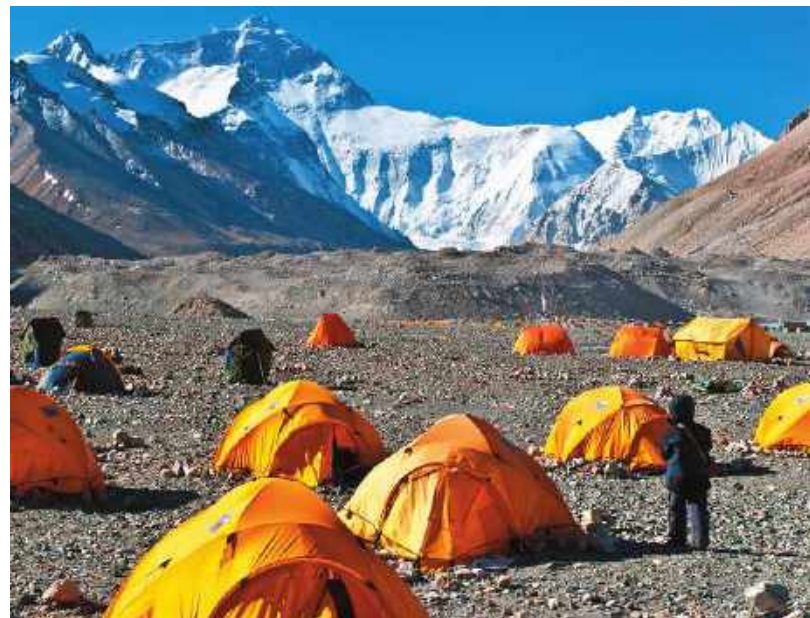


(b) The human circadian clock. Metabolic activities undergo daily cycles in response to the circadian clock. As illustrated for a typical individual who rises early in the morning, eats lunch around noon, and sleeps at night, these cyclic changes occur throughout a 24-hour day.

clock is normally coordinated with the cycle of light and darkness in the environment (see Figure 40.9b). For example, the hormone melatonin is secreted at night, and more is released during the longer nights of winter. External stimuli can reset the biological clock, but the effect is not immediate. That is why flying across several time zones results in jet lag, a mismatch between the circadian rhythm and local environment that persists until the clock fully resets.

Noting the importance of biological clocks to human health and disease, the Nobel Prize Committee awarded the 2017 Nobel Prize in Physiology or Medicine to Americans Jeffrey Hall, Michael Rosbash, and Michael Young, who studied the fruit fly *Drosophila* to map out the molecular mechanisms that underlie circadian rhythms.

▼ **Figure 40.10 Acclimatization by mountain climbers in the Himalayas.** To lessen the risk of altitude sickness when ascending a high peak, climbers acclimatize by camping partway up the mountain. Spending time at an intermediate altitude allows the circulatory and respiratory systems to become more efficient in capturing and distributing oxygen at a lower concentration.



Homeostasis is sometimes altered by **acclimatization**, an animal's physiological adjustment to changes in its external environment. For instance, when an elk moves up into the mountains from sea level, the lower oxygen concentration in the high mountain air stimulates the animal to breathe more rapidly and deeply. As a result, more CO_2 is lost through exhalation, raising blood pH above its normal range. As the animal acclimatizes over several days, changes in kidney function cause it to excrete urine that is more alkaline, returning blood pH to its normal range. Other mammals, including humans, are also capable of acclimatizing to dramatic altitude changes (**Figure 40.10**), although health risks remain.

CONCEPT CHECK 40.2

- 1. MAKE CONNECTIONS** How does negative feedback in thermoregulation differ from feedback inhibition in an enzyme-catalyzed biosynthetic process (see Figure 8.21)?
- 2.** If you were deciding where to put the thermostat in a house, what factors would govern your decision? How do these factors relate to the fact that many homeostatic control sensors in humans are located in the brain?
- 3. MAKE CONNECTIONS** Like animals, cyanobacteria have a circadian rhythm. By analyzing the genes that maintain biological clocks, scientists concluded that the 24-hour rhythms of humans and cyanobacteria reflect convergent evolution (see Concept 26.2). What evidence would have supported this conclusion? Explain.

For suggested answers, see Appendix A.